NEGOTIATING THE NILE: GAME THEORY, AI, AND DATA SOVEREIGNTY IN TRANSBOUNDARY WATER DIPLOMACY

Candidate Number: 4781

Word count (including footnotes, excluding title page and bibliography): 6997

1 Introduction: The New Age of Sovereignty in Water Disputes

According to the UN, managing transboundary water resources is projected to be one of the most pressing challenges facing human development over the next decade (UN Water, n.d.).[1] As the largest river in the world, supplying over 400 million people with water across Northeastern Africa, the Nile River has been a source of contention between Egypt, Sudan, and Ethiopia for decades (OWP, 2020). The recent construction of Ethiopia's Renaissance Dam (GERD) project has further exacerbated regional tensions. Consequently, discourse about the Nile River increasingly frames sovereignty not only as control over the resource itself, but also as an intertwined reflection of national identity and geopolitical influence. Trilateral negotiations to codify rights to the Nile's water have repeatedly stalled in an acrimonious deadlock, highlighting the urgent need for innovative strategies to establish a dynamic yet durable diplomatic framework for regional water cooperation (Yost, n.d.).

As a mathematical and conceptual tool, game theory provides a guiding structure for understanding power dynamics and strategic interactions, offering insights that can be applied to the development of effective diplomatic strategies. Moreover, the integration of Artificial Intelligence (AI) and Reinforcement Learning (RL) in this realm has refined analytic and predictive capabilities using computational simulations and real-time data mapping (Marwala, 2023). Although scholars argue for the potential of game-theoretic AI models to support cooperative mediation in transboundary water conflicts, their real-world effectiveness may be constrained by adversarial assertions of riparian sovereignty. AI-driven game theory models stand to introduce another dimension to the conflict, specifically data sovereignty, raising questions about how to design AI-negotiation tools that harmonize technical methodologies with complex geopolitical conflicts.

[1] Transboundary water disputes arise when states sharing water resources have conflicting interests (Ahuja, 2025).

This paper is anchored by the following research questions: Can AI-driven game theory tools offer meaningful support for diplomatic negotiations over transboundary water resources, such as the Nile? And if so, what challenges arise when sovereignty is not only a political principle but also embedded in data, models, and technological infrastructures? These questions are relevant given the shifting geopolitical landscape in Nile Basin, the acceleration of climate change, and the increasing use of digital technologies in diplomacy and governance. As the ongoing hydro-political conflict hinders development in the Nile Basin, this paper argues that novel computational negotiation methods — especially those combining game theory and AI — must be critically examined for their ethical implications and diplomatic viability. These technologies have potential to facilitate cooperative conflict resolution; but risk exacerbating existing power asymmetries if deployed without thoughtful design and adequate oversight. As such, this paper adopts an applied critical approach, using AI-enhanced game theory not as a prescriptive solution, but as a framework to evaluate both the opportunities and limitations of technological interventions in deeply political resource disputes.

As such, Section 2 provides the historical and political context of the Nile dispute, highlighting how sovereignty has shaped and constrained cooperative resolutions, with particular attention to the GERD's role in intensifying tensions. Section 3 reviews game theory, while Section 4 explores AI game theory applications in Nile negotiations, including multi-agent reinforcement learning models. Section 5 addresses data sovereignty in AI negotiation tools, as well as the ethical and technical limitations of AI modeling. Section 6 concludes with AI's broader implications in diplomacy and conflict management.

2 Political Context of the Nile Dispute

The Nile River, shown in Fig 1, is a vital water source in Eastern Africa, flowing 6,7000 kilometers through eleven countries and supplying over 250 million people with daily water (Zedan, 2018). With this population projected to double by 2050, amidst increasing flow variability and severe weather, the river is under growing strain (OWP, 2020). The Eastern Nile Basin, a large sub-basin draining the Nile and its tributaries, has become a particularly contested site for hydropower generation and potable water between Egypt, Sudan, and Ethiopia (Diriba & Li, 2021).[2] For over a century, these riparian states have been embroiled in a persistent political, legal, and doctrinal battle over sovereignty, all while attempting to reach a basin-wide agreement on the river's water allocation. In the last decade, Ethiopia's GERD reignited regional disputes over Nile rights by altering historical water allocation patterns and challenging Egypt and Sudan's longstanding control over the river (Bulto, 2009).

Figure 1: Nile Basin & GERD map (Ding et al., 2016)

While some scholars prognosticate competition over transboundary watercourses may provoke significant conflict, research shows cooperative water-sharing arrange-

[2]The Eastern Nile Basin is referred to as the Nile Basin (Bulto, 2009).

ments minimizes the potential for discord, and are usually achieved through a suite of strategic negotiation and bargaining tactics rather than armed conflict (Rieu-Clarke et al., 2012; Zeitoun & Warner, 2006). Though fears of large-scale water warfare are generally overestimated, unresolved conflicts over water can still severely curtail development; in the Nile Basin, this conflict impedes economic growth and increases regional instability (Wu & Whittington, 2006).

The GERD, and Basin conflict at large, illustrate a complex interplay between development goals, environmental stability, regional water security and sovereign claims (Almesafri et al., 2024). Although the construction and filling of the GERD has catalyzed conflict to date, it dually offers opportunities for shared benefits by providing stakeholders with potable water, flood control, and hydropower (Almesafri et al., 2024). Yet without a suitable framework for joint management and equitable allocation processes, the Nile remains at risk to irreversible overuse and the Basin's inhabitants face risks of acute water shortages, rising food insecurity, and escalating socioeconomic instability (Bulto, 2009). In this vein, effective negotiations over riparian governance and water allocation are quintessential to advance regional security, economic development and environmental sustainability (Yimer, 2015). However, the negotiation needs and objectives of Egypt, Sudan, and Ethiopia exemplify positional axioms steeped in deep-rooted historical and normative traditions which obstruct the scope for compromise or cooperation (Maru, 2020). As Zeitoun and Warner illustrate, transboundary water disputes are not solely about water but about safeguarding sentiments of state sovereignty (Zeitoun & Warner, 2006). These conflicts are deeply imbued with geopolitical power dynamics and national identity concerns (Zeitoun & Warner, 2006). Though cooperative water-sharing in the Nile Basin would yield mutual benefits for Egypt, Sudan, and Ethiopia, longstanding perceptions of power and control transcend notions of water parity, complicating negotiations. Sovereignty remains a core and contentious principle permeating the region's diplomatic landscape.

2.1 Defining Sovereignty

Sovereignty is a contested concept with various interpretations across disciplines and historical periods — from Jean Bodin's classical theory of state authority to contemporary understandings linked to territory, resources, and identity (Benoist, 1999). In this paper, sovereignty is understood as a state's power to govern itself, particularly in relation to its authority over natural resources and its autonomy in political and strategic decision-making (Pohle & Thiel, 2020). Within the Nile Basin, this manifests in states' claims to control and manage the river as an extension of their right to self-determination (Gümplová, 2020). However, in the age of AI, sovereignty now can extend into the digital sphere, where control over data — its collection, access, ownership, and use — constitutes a new form of geopolitical power referred to as data sovereignty. This shift introduces another layer of diplomatic complexity, particularly if algorithmic models help mediate negotiation and decision-making processes without clear frameworks for equitable data governance which will be discussed in Section 5.

In the Nile Basin, sovereignty is expressed through control over natural resources, particularly the river itself, which functions as a material asset and corollary to self-determination (Gümplová, 2020; McCreary & Lamb, 2014). While the dispute is rooted in competing claims to water access, it also reflects perceptions of historical power and national identity, making the Nile not just a resource, but a unique cornerstone of political legitimacy and regional dominance (Ayferam, 2023). As climate change intensifies and Ethiopia's GERD project reshapes the status quo, sovereignty has become existentially securitized, deepening mistrust and polarizing negotiation positions (Ayferam, 2023). In this complex environment, AI game theory technologies may offer means to bolster cooperative negotiations and mediate competing interests. However they also risk introducing new complications, particularly around data governance, perceived neutrality, and the digital extension of sovereignty claims if ethical and sovereign concerns are not preemptively considered (Srivastava & Bullock, 2024).

2.1.1 Physical Sovereignty and Resource Securitization

Emblematic of physical sovereignty, strategic control over the Nile has long been central to the foreign policies of riparian states as water security is essential for agriculture, energy, and transportation in the region (Osikena & David, 2010). Despite growing water scarcity, exacerbated by six major droughts in twenty years, no binding agreement exists to allocate the river's flow and antiquated colonial-era agreements remain as the status quo (IIJD, n.d.). Under these, Egypt claims about 66% of the Nile's annual flow, Sudan 22%, while the remaining water — subject to natural variability — must serve nine other nations, many of which are among the world's poorest, leaving these populations even more vulnerable to conflict, food insecurity, and displacement (World Bank, 2019).

The GERD has served as a lever for intensified tensions in the region.[3] While the dam will supply millions of households with electricity and expand national development prospects, it alters the downstream river flow to Egypt and Sudan (Kandeel, 2020). As such, Egypt, who is dependent on the Nile for 90% of its water has framed the dam as a direct threat to its resource sovereignty and petitioned the UN to denounce Ethiopia's behavior (Strasser, 2024; Yibeltal, 2024). In response, Ethiopia accused Egypt of perpetuating its "self-claimed monopoly" over the river (Yibeltal, 2024). Sudan has also expressed concern that the GERD will destabilize its water access and jeopardize its agricultural and energy sectors, though its position oscillates with shifting political leadership (Nigatu & Lidetie, 2025). Egypt's UN appeal and Ethiopia's accusation of hegemonic control illustrate how sovereignty over the Nile has moved beyond physical infrastructure to encompass symbolic and legal dimensions (Ayferam, 2023). This intensifying dispute sets the stage for the following analysis — exploring whether AI-enhanced game theory models can act as new diplomatic tools or risk deepening contestation over both physical and digital domains.

[3] Construction of GERD began in 2011

2.1.2 Historical Sovereignty and National Identity Politics

The GERD is a key factor in the Nile dispute, disrupting Egypt and Sudan's historically privileged water access and challenging national sovereignty norms. As Gümplová argues, control over natural resources has evolved into a core prerogative connected to state sovereignty and stability, linking jurisdiction over the Nile to broad conceptions of economic security and national development (Gümplová, 2020; Deribe, 2022). This is especially pertinent for Ethiopia, whose GERD represents a reclamation of national agency to advance energy and development goals (Atinafu, 2024). Conversely, Egypt and Sudan have discursively criticized the GERD, framing Ethiopia's pursuit to equalize its water share as a fundamental threat to the diplomatic order espoused in colonial-era treaties (Ayferam, 2023; Yimer, 2015).

Prior to the 20th century, water usage was largely informal and shaped by geographic constraints: Egypt and Sudan relied heavily on the Nile for agriculture, while Ethiopia, though upstream, lacked infrastructure to do the same (Trucker, n.d.). The 1929 Nile Waters Agreement between Egypt and Sudan, brokered by Britain, granted Egypt significant control over the Nile and formalized its "historic rights" to the river (Whittington, 2024). A subsequent 1959 Anglo-treaty further built upon this legislation, quantifying formal water allocations to be divided entirely between Egypt and Sudan (Swain, 2011). Ethiopia, was excluded, despite 85% of the Nile flow originating in its territory (Swain, 2011). Scholars such as Nagheeby and Amezaga critique these treaties for marginalizing upstream states and perpetuating colonial power structures, which Egypt and Sudan continue to invoke to justify their dominance (Nagheeby & Amezaga, 2023; Zedan, 2018).

This asymmetric legal legacy has entrenched a strategic yet ethically fraught dynamic in which control over the Nile equates to identity politics and power (Nagheeby & Amezaga, 2023). Though Nagheeby and Amezaga focus primarily on decolonizing water diplomacy, their work complements both Ayferam and Zeitoun & Warner who examine the confluence of resource sovereignty norms, hydropolitics, and national identity. For example, Ayferam posits that Egypt's national identity is centered on the ancient history of its Nile civilization (Ayferam, 2023). Similarly, the Nile is a unifying symbol for Sudanese heritage, serving as a lifeline for trade, cultural development, and trade (RA, 2023). Zeitoun and Warner provide a compelling framework illustrating that concerns over national identity are pervasive in negotiations over transboundary water, transforming the water's control into a proxy for legitimacy and authority (Zeitoun & Warner, 2006). Ultimately, the dispute over the GERD and river at large is about resource security as well as ontological security, or the preservation of state identity and geopolitical standing (Ayferam, 2023). Thus prescriptive concerns about resource security amalgamate with a nexus of historic significance and identity claims, pushing negotiations towards a zero-sum game where the river becomes a contested symbol of sovereignty as much as a shared resource (Deribe, 2022).

3 Game Theory and AI: Untangling the Nile's Sovereignty Paradox

As discussed in Section 2, the Nile dispute embodies a complex conflict over material resources, historical grievances, and national identity, and competing national identity — each underpinned by assertions of sovereignty. This produces what can be termed a sovereignty paradox: the more forcefully each state asserts its sovereign rights over the Nile, the more difficult it becomes to achieve the collective governance required to sustain the river as a shared resource (Ayferam, 2023). Navigating this paradox requires efficacious political negotiations, but also tools that can capture and formalize both strategic behavior and interdependence.

One such tool is game theory, which offers a structured framework for analyzing how actors with competing interests make decisions, negotiate, and potentially reach cooperative outcomes, often under conditions of uncertainty and mistrust (Mostafaei et al., 2025). In recent years, researchers have increasingly applied game-theoretic models to transboundary water conflicts (Tayia, 2019). In the context of the Nile dispute, game theory helps reveal how structural incentives and long-established preferences inhibit cooperation, even when mutual benefit is possible. So by applying game-theoretic logic, it becomes possible to move beyond normative calls for cooperation and instead model how cooperation could be achieved, or why it fails, by simulating strategic choices (Mostafaei et al., 2025). Scholars note game theory can provide the conceptual groundwork for more harmonious resource distribution models, which can steer states away from a "winner-takes-all" dynamic in the Nile Basin (Nigatu & Lidetie, 2025).

So as GERD negotiations continue, game theory can provide a structured framework to model strategic interactions and demonstrate how collective gains such as reduced evaporation loss through upstream storage and downstream access to hydropower can be achieved through coordinated actions that remain elusive to Nile Basin (Nigatu & Lidetie, 2025; Wu & Whittington, 2006). These models help untangle the sovereignty paradox at the heart of the dispute, where unilateral assertions of control undermine the shared governance required for regional stability. By simulating various negotiation scenarios, game theory offers insight into how states might shift from deep-rooted historical and political positions towards mutually advantageous strategies — moving from a zero-sum conflict towards a win-win scenario for water sharing in the Basin.

Furthermore, the integration of AI into game theory significantly enhances this potential, enabling dynamic modeling of complex geopolitical and environmental variables in real time (Marwala, 2023). AI-enhanced game theory models can simulate adaptive negotiation strategies and optimize water allocations based on environmental variability. AI game theory models also hold promising potential to incorporate both material and symbolic dimensions of sovereignty into simulations by encoding state-specific historical narratives, identity markers, and priorities into agent reward functions or scenario constraints, thereby modeling how non-material factors influence strategic behavior (Seyidova & Gojayev, 2023). Yet, the effectiveness of these models rests on their ability to engage with the historical grievances, geopolitical rivalries, and power asymmetries embedded in the conflict. Crucially, questions of data sovereignty — who controls, contributes to, and interprets the data used to train these models — further complicate their legitimacy, neutrality, and impact in real-world diplomacy, as will be discussed in Section 5. To address these ethical and technical concerns through a grounded analysis, the remainder of Section 3 outlines how game theory can model negotiation dynamics in the Basin. Section 4 will then show how AI game theory models can open new avenues for diplomatic conflict resolution using RL for negotiation simulations and modelling.

3.1 Game Theory: Modeling Negotiations in the Nile Dispute

Game theory offers a valuable framework for analyzing the Nile dispute, where states are driven by deep-rooted values of sovereignty and self-determination. Originally developed by John von Neumann and refined by John Nash, game theory is a branch of applied mathematics that models how rational actors, either states or individuals, make decisions in response to the anticipated actions of others who hold similar, opposed, or mixed interests (Garon, 2023; Ross, 2024). Game theory has been widely used in resource management, providing a structured approach to explore optimal strategies for cooperative water-sharing (Dinar & Hogarth, 2015). Compared to conventional economic models, game theory is particularly well-suited to transboundary water conflicts for several reasons: it maps conflict resolution processes, accommodates multiple decision-making criteria, and captures the interactive dynamics between diverse individual and state entities (Tayia, 2019). Both non-cooperative and cooperative game theory models have been applied to water conflicts, yielding insights into potential outcomes and pathways for diplomatic resolution (Mostafaei et al., 2025).

Non-cooperative Dynamics of Dispute

In game-theoretic terms, the Nile dispute resembles a multi-player game; each country seeks to maximize their own utility (e.g. water for agriculture or hydropower) often at the expense of others. By definition, players in noncooperative games make decisions without formal contracts, aiming to protect their own interests without considering the group's welfare (Nash, 1951). So without binding agreements, each state is incentivized to adopt a strategy that prioritizes sovereign objectives; for example, Ethiopia may impound as much water as possible while Egypt may threaten legal or martial action to deter any reduction in its water share (Almesafri et al., 2024). Sudan, situated both geographically and diplomatically between the two, must balance its water needs with shifting regional alliances related to upstream GERD developments (Almesafri et al., 2024). Thus far, Egypt, Ethiopia, and Sudan have pursued more unilateral moves, including uncoordinated dam filling and independent water extractions.

This dynamic closely mirrors the Prisoner's Dilemma, a classic game theory scenario in which rational self-interest leads to mutual defection and suboptimal outcomes instead of cooperation with the other party (Evans, 2023).[4] The resulting Nash Equilibrium — a situation where no actor can improve their outcome by changing their own strategy in relation to others (Evans, 2023). In the Nile context, the resulting Nash Equilibrium leads to collective defection, leading to increased water scarcity, instability, and conflict. Deep mistrust and sovereignty concerns reinforce this deadlock, exemplifying the "sovereignty paradox" where efforts to assert control undermines the potential for collective well-being and equitable cooperation (The Sovereignty Paradox, 2013). As a non-cooperative game lacking strong external enforcement mechanisms to cultivate trust, assure sovereign claims, or satisfy all

[4]The Prisoner's Dilemma originates from a hypothetical scenario where two prisoners face the choice to betray each other or stay silent, showing conflict of individual self-interest and collective benefit (Evans, 2023).

parties' needs, the Nile negotiations have repeatedly failed (Kairon et al., 2020). Overall, non-cooperative game theory thus provides a useful lens for understanding the strategic rigidity of each state's preferences and options, shaped by both material interests and identity-based sovereignty claims. Fig 2 illustrates the Prisoner's Dilemma in the Nile Dispute, with trilateral defection equating to the current status quo; the more each country prioritizes its own interests over cooperation, the worse the overall outcome becomes for everyone involved.

Noncooperative game theory has been widely applied to transboundary water conflicts, such as disputes over the Helmand River and Great Lakes (Tayia, 2019). In Iran, Nazari et al. used noncooperative game dynamics to create a simulation-optimization model to equalize water distribution, while Mehrparvar et al. applied similar methods to resolve allocation disputes in the Zayandehroud Basin (Mehrparvar et al., 2020; Nazari et al., 2020). Similarly, Madani developed a non-cooperative game-based decision support system to help predict strategic behavior in the Nile Basin, though it has been criticized for oversimplifying the sociopolitical dynamics of the Basin (Madani, 2010). Moreover, Agrawal and Jaiswal proposed an algorithm to circumvent the typical Prisoners Dilemma scenario, demonstrating that machine learning can potentially curtail mistrust and unilateralism that arise in noncooperative multi-player negotiations (Agrawal & Jaiswal, n.d.). Although noncooperative game theory proves advantageous for analyzing the strategic behavior of inherently self-interested states, many scholars argue cooperative game theory provides a valuable frameworks crucial to managing shared resources (Mirzaei-Nodoushan et al., 2022) .

[5] Bilateral defection results in a 4, as multiple defectors dilute gains.

# Defectors	Egypt (A)	Ethiopia (B)	Sudan (C)	Payoffs (A, B, C)	Outcome
0	C	C	C	(3, 3, 3)	Trilateral Cooperation
1	C	C	D	(1, 1, 5)	Sudan defects, ***gains***
1	C	D	C	(1, 5, 1)	Ethiopia defects, ***gains***
1	D	C	C	(5, 1, 1)	Egypt defects, ***gains***
2	C	D	D	(0, 4, 0)	Bilateral defection
2	D	C	D	(0, 0, 4)	Bilateral defection
2	D	D	C	(4, 0, 0)	Bilateral defection
3	D	D	D	(1, 1, 1)	Trilateral defection

Key

- 0 Defectors – Trilateral cooperation
- 1 Defector – Temptation to defect
- 2 Defectors – Mutual loss begins
- 3 Defectors – Trilateral defection

Payoff Scale

- 5 Highest gain from defection while others cooperate
- 3 Mutual cooperation
- 1 Mutual defection *(minimal return)*
- 0 Cooperating while others defect

Figure 2: Example Three-Party Prisoners' Dilemma Game [5]

Cooperative Game Theory and Benefit-Sharing

Cooperative game theory explores how actors can form coalitions to achieve mutually beneficial outcomes (Peleg & Sudhölter, 2007). Hossen et al.'s review of global river disputes highlights benefit-sharing as the most effective approach to resolving transboundary water conflict, as it moves a zero-sum allocation problem into a win-win scenario (Hossen et al., 2023). Research shows that many interactions over transboundary resources have been cooperative (67%), with over 650 cooperative international agreements signed, based on findings from the Water International Journal (Diplomat, 2024). As Tayia notes, cooperative game theory has been successfully operationalized in the Ganges River, and major basins in Columbia and Jordan

(Tayia, 2019). Similarly, McKinney implemented a trilateral cooperative game in the Syr Darya Basin to sequentially model water allocations, improving riparian negotiations and optimal resource sharing (McKinney, 2007). So too, Tharwat and Sabry developed allocation models for Iran's Rafsanjan Basin, emphasizing the merit of cooperative game theory to "maximize the total net benefit of shared stakeholders" (Tharwat et al., 2024).

In the Nile Basin, a cooperative arrangement might involve benefit-sharing mechanism: for example, Ethiopia could guarantee minimum downstream flows and affordable electricity exports to Egypt and Sudan in exchange for compensation or development assistance. This mirrors a game theory Nash Bargaining solution (NBS), which outlines how to efficiently divide resources by maximizing shared utility while requiring each party to make acceptable concessions (Binmore et al., 1986). NBS models have seen success in real-world applications: a recent two-stage model in the Tigris–Euphrates Basin utilized NBS to adjust water allocations to stakeholders based on risk and necessity considerations (Wu et al., 2022). A NBS model rooted in cooperative game theory may help deescalate negotiations by promoting more equitable outcomes for water divisions while in turn, respecting sovereignty, as exemplified by Janjua et al.'s collaborative model to improve inter-province stability in the Indus River Basin (Janjua et al., 2025). AI can enhance cooperative game theory NBS models further by simulating repeated bargaining scenarios for negotiations with real-time data that adjusts in response to changing political and environmental factors (Schelble et al., 2021). As such, AI game theory models may offer opportunities to craft more innovative water-sharing negotiation strategies tailored to the intricacies of the dispute. While the inextricable link between the Nile dispute and sovereignty complicates cooperation, AI game theory models designed to account for sovereign claims could complement negotiation processes and facilitate acceptable trilateral outcomes. In summation, game theory provides the mathematical foundation for modeling strategic interactions, while AI enhances these models with dynamic and predictive capabilities to optimize decision-making for competitive and cooperative solutions (Hazra et al., 2024). Notwithstanding the applicative benefits of such models, it's important to recognize that both approaches have limitations — particularly in oversimplifying diplomatic complexities and politicizing data — which Section 5 will explore (Colman, 2003).

4 AI Decision-Making in the Nile

Game theory is a useful tool to investigate muti-actor interactions in transboundary water disputes; AI technologies enhance these analytical frameworks by simulating negotiations, incorporating water management metrics from hydrological data, and calculating adaptive multi-objective outcomes (Haider et al., 2024). Several studies have demonstrated the effective convergence of game theory and AI to reduce conflict over water resources. For instance, Li et al. developed an AI negotiation protocol for China's South-to-North Water Transfer Project, using a game-theory bidding algorithm to ensure a more equitable regional water distribution (Li et al., 2014). In addition, Zeng et al. introduced a hybrid model combining game theory and mathematical programming to address transboundary water conflict in the Guanting basin, based on qualitative and aggregate quantitative factors (Zeng et al., 2019). As Sarkar and Jha note, AI can play a "critical role in equitable water management" and foster cooperative solutions using transparent and "impartial data for predictive modeling" (Sarkar & Jha, 2025).

AI game theory models can support negotiations between Egypt, Sudan, and Ethiopia by providing insights on reservoir operations, irrigation scheduling, and infrastructure management to better divide water across the countries while minimizing waste (Sarkar & Jha, 2025). By integrating hydrological, economic, and political data, AI game theory systems can model cooperative water-sharing parameters tailored to the region's unique needs (Zomorodian et al., 2017). While wholly depoliticizing such deeply entrenched resource disputes may be unrealistic, studies suggest AI game theory models can reframe negotiations by shifting focus toward optimal outcomes and factual scenario analysis, thereby creating a more objective basis for discourse (Seyidova & Gojayev, 2023). In one such promising case, Woods combined normative ethics with game-theoretic "cake-cutting" models to address the Nile dispute, demonstrating AI algorithms can offer practical pathways to balance resource asymmetries and support equitable divisions (Woods, 2024).[6] Fig 3 includes several examples of transboundary water studies where game theory and AI systems have dovetailed effectively, yielding valuable insights into strategic decision-making and cooperative resource management.

Beyond resource allocation, further studies reveal AI is recognized is emerging as a tool for conflict resolution more broadly, fostering understanding, enhancing empathy, and facilitating proactive solutions through the integration of analytics (Albrecht, 2023; The Impact of Technology on Conflict Resolution, 2023). While AI-integrated game theory can help simulate cooperative solutioning and offer practical insights to mitigate conflict over scarce resources, they often rely on fixed strategies and assumptions about rational behavior. Real-world diplomacy is dynamic: states continuously adjust their positions based on others' actions and shifting environmental conditions. Likewise, incorporating the issue of sovereignty into negotiations models may require advanced computing techniques to capture evolving political and national interests. To address this complexity, recent research has turned to multi-agent reinforcement learning (MARL), a subfield of AI, as a more adaptive AI framework that builds on game theory while allowing agents to learn from experience over time and adapt strategy. In the Nile Basin, a MARL may be a viable to model negotiation dynamics more realistically, enabling simulations that reflect the shifting, conflicting geopolitical interests.

[6] Cake cutting is a mathematical method for dividing resources fairly among participants (Woods, 2024).

<table>
<tr><th>AI Capability</th><th>Benefit</th><th>Case Study</th></tr>
<tr><td>Hydrological Modeling & Forecasting</td><td>Produce more accurate multistep flow forecasts
Help parties anticipate rain/drought conditions to reduce uncertainty in negotiations over annual water shares</td><td>AI management program to optimize water resource allocation in Zhoushan Islands, China (Guo et al., 2021)
AI analysis to identify hydrological patterns in Narmada River Basin, India (Vassar Labs, 2024)</td></tr>
<tr><td>Real-time Adaptive Management</td><td>Process real-time data and adjust operations dynamically (e.g. dam releases), like “smart grid” controls in energy systems
Minimize waste in response to changing conditions</td><td>AI algorithm to schedule water allocations and minimize water loss based on real-time data (Hu et al., 2023)
Adaptive AI framework to address water scarcity in South Africa (JDG Frame et al., 2018)</td></tr>
<tr><td>Decision Support and Scenario Analysis</td><td>Rapidly simulates thousands of water-use scenarios to reduce waste and optimize allocations
Can be used for stress-testing proposals and finding compromise</td><td>AI-based framework for multi-reservoir system used stochastic inflow to account for uncertainty in Segura Basin, Spain (Otamendi et al., 2024)
AI support system for water management (Zhao et al., 2025)</td></tr>
</table>

Figure 3: Applications of AI in Water Resource Management

4.1 MARL Models: Dynamic Adaptation in Nile Negotiations

MARL extends AI-decision-making and game theory into environments with multiple autonomous agents, such as countries, each pursue their own goals. Unlike static models, MARL simulates dynamic interactions, allowing agents (e.g. water users) to adapt strategies in response to changing conditions and one another's behavior. This makes MARL well-suited to analyze intricate diplomatic situations involving numerous actors with diverse preferences and abilities, which is often the case in sequential water management decisions (Amigoni et al., n.d.).

Recent research highlights MARL's potential to optimize water allocations in competitive, multi-stakeholder settings. For example, Ding et al. suggested a framework similar to a MARL to allocate Nile water to eleven countries, creating a redistribution mechanism accounting for each agent's self-interest while promoting fairness (Ding et al., 2016). Hung et al. also simulated dynamic human-nature interactions with a RL agent-based model in the Colorado River Basin, showing agents could learn to adjust water demands over time, even in the face of stochastic unknowns and system errors (Hung et al., 2021). Additionally, a joint study by Doukkali University and the Centre d'Etudes Spatiales ran computer simulations mimicking multi-agent negotiations between water suppliers and farmers to share limited water resources, demonstrating that adaptive strategies outperformed static allocation models in terms of efficiency (Belaqziz, 2016). These examples highlight how MARL enables negotiation-based mechanisms that adapt to real-time rules, iteratively refine trade-offs, and guide compromise based on equative logics. In the context of the Nile dispute, the MARL can train AI agents, representing each country, to negotiate and coordinate water-sharing policies through repeated interactions in a simulated environment, as shown in Fig 4.

Here, a MARL approach marries elements of game theory (multiple utility-seeking actors) with AI machine learning as agents improving their strategies over time (Alonso & Njupoun, 2024). Additionally, a MARL system could incorporate sovereignty interests into the model by assigning distinct utility functions to each riparian state that reflect their unique priorities, such as water rights, economic needs, and or historical claims (Haider et al., 2024). Through iterative learning and repeated simulations, the system could learn how these priorities interact and identify cooperative strategies that balance these claims while promoting more balanced and diplomatically feasible solutions.

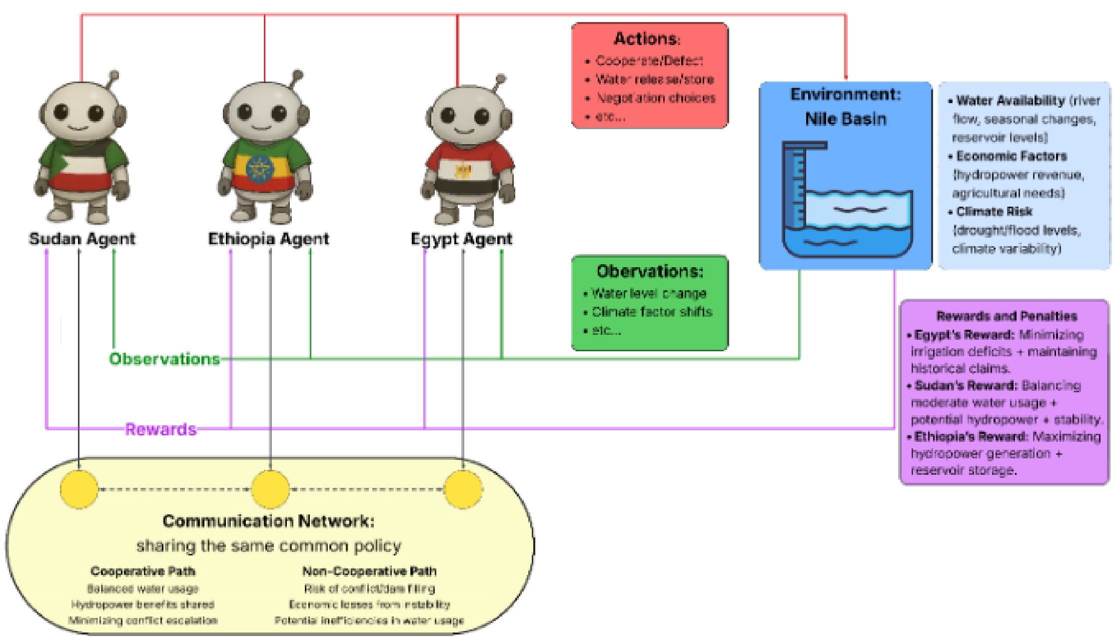

Figure 4: Example Nile Dispute MARL

4.1.1 Applications for MARL in the Nile Dispute

Negotiation Simulations

MARL presents a promising framework to overcome the protracted stalemate that has stalled cooperation in the Nile Basin by modeling how negotiation dynamics evolve over time. By simulating repeated interactions between autonomous agents each representing a state with its own interests in a virtual "laboratory," MARL can explore how strategy changes over time with changing conditions. This flexibility is essential in the Nile context, where sovereignty claims are not only historically ingrained but also shaped by shifting leadership, environmental pressures, and regional alliances. A system capable of learning and adapting alongside these changes offers a valuable counterpoint to more traditional, static modeling approaches.

Each AI agent in a MARL simulation can be programmed with a reward function reflecting the specific priorities of its respective country — such as Ethiopia's focus on hydropower, Egypt's emphasis on uninterrupted water flow, or Sudan's interest in balancing both. As the agents interact, they learn which actions lead to favorable outcomes, adjusting strategies in response to others' behavior (Alonso & Njupoun, 2024). Rather than assuming perfectly rational behavior, the agents learn which strategies produce favorable outcomes over time through trial and error. This enables the model to approximate diplomatic processes including trust-building, strategic retaliation, compromise, and shifting alliances — dimensions often absent in current negotiation frameworks (Leibo et al., 2017). For instance, Ethiopia may learn that aggressive water consumption provokes retaliatory action by others and cooperative approaches can achieve more advantageous outcomes, as found with MARL irrigation studies in India where MARL policies helped farmers adopt adaptive agricultural practices (Mahajan, 2025).

While current research has proposed variations of MARL for the Nile Basin focusing on equitable revenue redistribution, such as Ding et al., much of its application remains prospective, especially when it comes to integrating complex sovereignty claims (Ding et al., 2016). However, related work demonstrates the future feasibility of this technology. For example, Hung et al. modeled adaptive water management in the Colorado River Basin, showing how agents can adjust behavior in response to hydrological and institutional feedback (Hung et al., 2021). Belaqziz reached similar conclusions in a study of agricultural water-sharing, demonstrating that adaptive negotiations can outperform static allocation schemes (Belaqziz, 2016). These studies suggest MARL can serve as dynamic tool for stress-testing different negotiation situations under complex, real-world conditions (Aponte-Rengifo et al., 2023).

Adaptive Treaties

Beyond simulating negotiations, MARL can also be applied to adaptive treaty frameworks that respond to environmental changes and real-time interactions between states (Mason et al., 2016). Transboundary water treaties are often rigid and outdated, which cannot reflect present-day realities of sovereignty and climate variability (Jafroudi, 2020). By running simulations that incorporate climate patterns, political shifts, and economic trends, MARL can adjust cooperative treaty policies. For example, a MARL can adapt to environmental fluctuations; agents can learn to temporarily reallocate more water to Sudan or Egypt during dry years, while ensuring Ethiopia's sovereign right to use the GERD is upheld through compensation mechanisms during stable water years (Yao et al., 2025). Kontak et al. recently demonstrated that single-agent RL can optimize Nile reservoir management across competing goals (e.g. irrigation quotas and hydropower production), and extending the model to MARL settings enables viable policy coordination; though achieving perfect equilibrium remains nebulous as managing a complex environment like a river basin can be difficult and requires extra training to guide the system's learning process (Kontak, 2024).

Overall, MARL simulations can provide valuable insights into the long-term impacts of policy choices and negotiation strategies to reach optimal outcomes in the

Nile Basin. In this, MARL can embed sovereignty-sensitive objectives into simulations if agents are programmed with parameters derived from socioeconomic, political, and historical data. By evaluating technical feasibility and political acceptability, MARL can support prototyping treaty designs that respect sovereignty while encouraging sustainable cooperation. As a technological bridge between the formalism of game theory and fluidity of real-world diplomacy, MARL would not replace human-led diplomatic negotiation but support them by offering space for testing adaptive governance strategies in advance.

5 Data Sovereignty and Limitations of AI-Powered Diplomatic Tools

While AI game theory technologies, including MARL models, offer unconventional approaches to simulate strategic negotiations, the application of AI diplomatic contexts raises critical concerns — ranging from their technical viability to the political implications of their design and deployment, all of which may reinforce existing power asymmetries. Traditional sovereignty in the Nile Basin has been contested through territorial control, historical narratives, and national security policies. AI game theory models introduce a new and often understudied dimension of geopolitical power: data sovereignty which can be understood as the right of a state to control the collection, ownership, and use of data generated within its borders (Calzada, 2025; Niklas, 2025). In AI-mediated negotiation models, this concept expands to include who defines the problem, controls the data inputs, interprets outputs, and shapes the boundaries of decision-making itself.

5.1 Data Sovereignty and Ethical Challenges

AI systems are presented as neutral, mediating tools by many scholars, but in high-stakes diplomatic contexts they can potentially perpetuate exclusionary dynamics and promote historical hierarchies of control. The issue of data sovereignty is not only about ownership, but about epistemic authority: whose realities, values, and futures are made legible within the system (Abbas et al., 2024). In the Nile Basin, where data-sharing is historically opaque and mistrust among riparian states is high, AI models risk reinforcing existing asymmetries if they are not designed collaboratively. The origin and control of data for AI models can become a renewed site of contention for sovereignty; in the absence of formal data infrastructure and agreed-upon protocols for inclusion, the model may privilege technical capacity, strategic interests, or ideological assumptions of one state over others (Kansara & Lakshmi, 2022).

For example, effective AI negotiation models would require access to real-time hydrological data including streamflow records, dam operations, rainfall forecasts, and irrigation demands (Haider et al., 2024). Without equitable data-sharing frameworks or transparency metrics for sourcing and validating data, inaccurate or biased data — rooted in disproportionate colonial histories favoring downstream nations — can skew model outputs (Source). Alternatively, Ethiopia's control over data from the GERD may afford it strategic advantage in shaping predictive AI outputs, while Egypt and Sudan may be left dependent on third-party satellite monitoring or datasets that do not reflect local realities. Even if external organizations like the UN FAO or World Bank provide this data, they bring their own institutional legacies and situational assumptions, which can subtly shape the neutrality of the model and its perceived legitimacy (Swain, 2011). While regional organizations like the African Union are better positioned to oversee AI model development with contextual sensitivity, they still may lack the technical capacity to supply comprehensive, unbiased data (Yilma, 2023).

This has broader implications. If the structure of the AI model is treated as apolitical, but does embed specific configurations of sovereign power, it risks reproducing colonial, extractive, or technocratic logics under the guise of objectivity. In this way, AI models themselves become sites of contestation over resources, representation, and governance. Data sovereignty extends beyond the politics of infrastructure to how the model is created, whose data it uses, and how its outputs are validated, therefore co-creation of AI models is essential. All states involved must participate in setting the model's objectives, defining constraints, verifying the integrity of shared data (Sarkar & Jha, 2025). Sovereignty should be treated as a principle that guides the architecture of the mode, not as a fixed input in the system. Though the states may co-create the model to reflect their own interests, particularly in setting objectives and acceptable trade-offs, the model itself should be structured to evaluate inputs against shared, transparent criteria to create procedural fairness and "quantify" sovereign interests to a degree. Collaboratively curating model data requires mutual trust and cooperation and may be unlikely unless sovereignty can serve as a "key organizational principle" to promote environmental conservation and sustainable water, as Schrijver notes (Schrijver, 2021). As the World Economic Forum cautions, AI cannot substitute for human judgment in geopolitical decisions and its autonomous nature may provoke resistance if perceived as undermining national agency (WFE, 2024). Moreover, the risk of opaque AI models with biased data and limited interpretability pose technical challenges with political ramifications that can erode trust, legitimacy, and diplomacy.

5.2 Technical Modeling Challenges

In general, AI models face challenges including limited real-world accuracy, goal misalignment, and inherent difficulties testing performance in sensitive diplomatic contexts (Caballar, 2024). Notwithstanding these concerns, MARL still holds promise for simulating cooperation strategies in the Nile Basin, but its effectiveness also depends on high-quality reliable data, which remains difficult to obtain in transboundary river environments. Hydrological modeling of the Nile is inherently complex due to remote headwaters, seasonal variability, and limited monitoring infrastructure; measurement stations have declined for the past three decades, complicating states' abilities to generate accurate water level predictions (Melesse, 2024; Tarhule & Hasan, 2020). As Abtew et al. emphasize, inconsistencies in measurement methods make standardization even more critical in negotiation tools like MARL, which rely on accurate inputs (Abtew et al., 2021). Reliable hydrological data is crucial for powering MARL systems and producing actionable insights. With regional tensions already high in the Nile Basin, perceptions of misclassified, biased, or unreliable data feeding AI negotiation instruments could destabilize negotiations or catalyze conflict further.

Hydrological data is not just a technical consideration, but political. In the Mekong Basin, erroneous dam data led to accusations that China was withholding water from downstream nations, heightening discord and mistrust (Fudan Institute, 2022). In the Helmand River dispute, Loodin et al. shows how data sharing between Afghanistan and Iran remains constrained by national security concerns, mistrust, and competing sovereignties (Loodin et al., 2023). As seen with the Helmand case, water data is not solely a technical input, but a strategic asset, and transforms sovereignty along new technological dimensions. Even when technical capacity exists, data can still be withheld and contested in mediations, reinforcing power hierarchies rather than reducing them. These cases illustrate that data sovereignty — control over where and how data is produced and used — shapes the legitimacy of AI models in diplomatic negotiations. Without transparent data governance and mutual trust, MARL systems risk encoding existing inequalities and further politicizing resource disputes. For AI to support cooperation in the Nile Basin, shared infrastructure and co-developed data protocols must be treated not as technical add-ons, but as central to the model's credibility.

6 Conclusion: Nile Dispute, AI Technology, and Diplomacy

This paper examines how AI-driven game theory models can potentially facilitate negotiations and adaptive compromise in the Nile dispute, while highlighting the ethical and technical considerations necessary to ensure the design and deployment of these models navigate critical contestations of sovereignty. The Nile dispute illustrates how sovereignty over shared resources remains intertwined with a nexus of identity, security, and power. To untangle these complexities, AI game theory, and MARL models in particular, offer promising tools for simulating negotiations and exploring cooperative diplomatic outcomes with unprecedented precision (Pasupuleti, 2024). However, their effectiveness depends not only on technical sophistication, but also on inclusive design, transparency data sharing, and sensitivity to historical and geopolitical asymmetries. These AI tools should not replace human diplomacy but reimagine it - supporting more informed and adaptive approaches to conflict resolution and governance. Likewise, proactively managing data sovereignty through shared parameters and inclusive design is key to preventing AI models from becoming another source of geopolitical contention.

In conclusion, without careful attention to data sovereignty and fairness, such AI technologies risk replicating existing hierarchies rather than resolving them. AI game theory models should function as neutral decision-support tools, helping the states simulate trade-offs and explore equitable outcomes grounded in common environmental realities, rather than contested political narratives. Still, technically sound AI models may face limits in sensitive political settings, where their legitimacy depends on being framed as aids to human judgment, not automated enforcers. Furthermore, unresolved questions around data access, ownership, predictive reliability, along with cost, institutional capacity, and societal acceptance remain critical ethical concerns in deploying such models in the Nile Basin specifically. As such, further research is needed to understand how to feasibly design these tools to appropriately reflect real-world diplomatic complexities, wholly account for sovereign interests, and accrue reliable data to train models.

Nonetheless, the integration of AI into diplomacy reflects a broader shift in how power relations are mediated through technological systems, introducing new actors, tools, and domains of influence (Mazrouei, 2025). The Nile dispute offers a lens to examine the broader confluence between AI technologies and conflict management;

as many scholars in Asia and the Middle East are already adopting RL and game theory methodologies for resource governance frameworks, challenging the notion that this innovation is solely driven by Western powers (Okolo, 2023). In this vein, Egypt, Ethiopia, and Sudan should pursue these technologies on their own terms, using AI tools to redefine sovereignty conceptions through mutually beneficial cooperation. Although focused on the Nile Basin, this analysis illustrates how historical geopolitics continue to shape sovereignty and resource conflicts — and as AI becomes increasingly embedded in diplomacy, control over data and decision-making will be as consequential as control over physical resources themselves.

Bibliography

Abbas, A. E., van Velzen, T., Ofe, H., van de Kaa, G., Zuiderwijk, A., & de Reuver, M. (2024). Beyond control over data: Conceptualizing data sovereignty from a social contract perspective. Electronic Markets, 34(1), 20. https://doi.org/10.1007/s12525-024-00695-2

Abtew, W., Moges, S. A., Meles, M., & Imru, M. (2021). River Flow Monitoring and Data Quality for Equitable Nile Water Sharing. In A. M. Melesse, W. Abtew, & S. A. Moges (Eds.), Nile and Grand Ethiopian Renaissance Dam: Past, Present and Future (pp. 295– 314). Springer International Publishing. https://doi.org/10.1007/978-3-030-76437-1_16

Agrawal, A., & Jaiswal, D. (n.d.). When Machine Learning Meets AI and Game Theory. Ahuja, K. (2025, February 8). Transboundary Water Disputes and International Legal Mechanisms. Bhatt & Joshi Associates. https://bhattandjoshiassociates.com/transboundary-water-disputes-and-internationallegal-mechanisms/

Al Mazrouei, N. (2025). AI-Powered Diplomacy: The Role of Artificial Intelligence in Global Conflict Resolution. https://trendsresearch.org/insight/ai-powered-diplomacy-the-role-ofartificial-intelligence-in-global-conflict-resolution/

Albrecht, E. (2023). Predictive Technologies in Conflict Prevention: Practical and Policy Considerations for the Multilateral System. Discussion Paper.

Almesafri, A., Abdulsattar, S., Alblooshi, A., Al-Juboori, R. A., Jephson, N., & Hilal, N. (2024a). Waters of Contention: The GERD and Its Impact on Nile Basin Cooperation and Conflict. Water, 16(15), Article 15. https://doi.org/10.3390/w16152174

Alonso, M., & Njupoun, A. (2024). Game Theory and Multi-Agent Reinforcement Learning: A Mathematical Overview (SSRN Scholarly Paper No. 4926191). Social Science Research Network. https://doi.org/10.2139/ssrn.4926191

Amigoni, F., Castelletti, A., & Gazzotti, P. (n.d.). Water Resources Systems Operations via Multiagent Negotiation (Extended Abstract).

Aponte-Rengifo, O., Vega, P., & Francisco, M. (2023). Deep Reinforcement Learning Agent for Negotiation in Multi-Agent Cooperative Distributed Predictive Control. Applied Sciences, 13(4), Article 4. https://doi.org/10.3390/app13042432

Atinafu, L. (2024). GERD: Powering Ethiopia's economic transformation – Ethiopian Press Agency. https://press.et/herald/?p=101673

Ayferam, G. (2023). The Nile Dispute: Beyond Water Security. Carnegie Endowment for International Peace. https://carnegieendowment.org/sada/2023/01/the-nile-disputebeyond-water-security?lang=en

Belaqziz, S. (2016). Simulating Negotiations over Limited Water Resources: A Multi-Agent System Approach for Irrigation Systems Facilitates the Analysis of the Decision-making Process.

Benoist, A. (1999). What is Sovereignty? https://crusoesisland.wordpress.com/wpcontent/uploads/2012/04/what-is-sovereignty-two-treatises-of-govt.pdf

Binmore, K., Rubinstein, A., & Wolinsky, A. (1986). The Nash Bargaining Solution in Economic Modelling. The RAND Journal of Economics, 17(2), 176–188. https://doi.org/10.2307/2555382

Bulto, T. S. (2009a). Between Ambivalence and Necessity: Occlusions on the Path Towards a Basin-Wide Treaty in the Nile Basin (SSRN Scholarly Paper No. 1469795). Social Science Research Network. https://papers.ssrn.com/abstract=1469795

Bulto, T. S. (2009b). Between Ambivalence and Necessity: Occlusions on the Path Towards a Basin-Wide Treaty in the Nile Basin (SSRN Scholarly Paper No. 1469795). Social Science Research Network. https://papers.ssrn.com/abstract=1469795

Caballar, R. (2024). 10 AI dangers and risks and how to manage them — IBM. https://www.ibm.com/think/insights/10-ai-dangers-and-risks-and-how-to-manage-them

Calzada, I. (2025). Decentralizing Power? Data Sovereignty in the Age of AI and Web3 (SSRN Scholarly Paper No. 5081504). Social Science Research Network. https://doi.org/10.2139/ssrn.5081504

Colman, A. M. (2003). Cooperation, psychological game theory, and limitations of rationality in social interaction. Behavioral and Brain Sciences, 26(2), 139–153. https://doi.org/10.1017/S0140525X03000050

Deribe, M. (2022, February 15). The 1959 Agreement "for the full utilization of the Nile waters": The crux of the problem in the Nile Basin water use - weASPiRE. https://www.weaspire.info/the-1959-agreement-for-the-full-utilization-of-the-nile-watersthe-crux-of-the-problem-in-the-nile-basin-water-use/

Dinar, A., & Hogarth, M. (2015). Game Theory and Water Resources: Critical Review of its Contributions, Progress and Remaining Challenges. Foundations and

Trends in Microeconomics, 11(1–2), 1–139. https://doi.org/10.1561/0700000066

Ding, N., Erfani, R., Mokhtar, H., & Erfani, T. (2016). Agent Based Modelling for Water Resource Allocation in the Transboundary Nile River. Water, 8(4), Article 4. https://doi.org/10.3390/w8040139

Diplomat, T. W. (2024, April 4). New article explores transboundary water conflict and cooperation trends. The Water Diplomat. https://www.waterdiplomat.org/story/2024/04/new-article-explores-transboundary-waterconflict-and-cooperation-trends

Diriba, H., & Li, F. (2021). Energy Sector Status and Hydropower Development in the Eastern Nile Basin. OALib, 08(04), 1–14. https://doi.org/10.4236/oalib.1107338

Evans, L. (2023). Understanding Nash Equilibrium: A Comprehensive Guide to Economic Models and Game Theory. Principlesofeconomics.Net. https://www.principlesofeconomics.net/game-theory-nash-equilibrium

Fudan Institute. (2022). How did false data turn the Mekong River into an 'issue'? - Chinese— Fudan Institute of Belt and Road & Global Governance. https://brgg.fudan.edu.cn/en/reports.html/webeditor/uploadfile/file/webeditor/uploadfile/file/articleinfo_968.html

Game theory — Definition, Facts, & Examples — Britannica. (2025, February 12). https://www.britannica.com/science/game-theory

Garon, C. (2023, February 9). Game Theory: Real-Life Examples and Applications. Christophe Garon. https://christophegaron.com/articles/mind/game-theory-real-life-examples-andapplications/ Gümplová, P. (2020a). Sovereignty over natural resources – A normative reinterpretation. Global Constitutionalism, 9(1), 7–37. https://doi.org/10.1017/S2045381719000224

Gümplová, P. (2020b). Sovereignty over natural resources — A normative reinterpretation. Global Constitutionalism, 9(1), 7–37. Scopus. https://doi.org/10.1017/S2045381719000224

Guo, Y., Yu, X., Xu, Y.-P., Chen, H., Gu, H., & Xie, J. (2021). AI-based techniques for multistep streamflow forecasts: Application for multi-objective reservoir operation optimization and performance assessment. Hydrology and Earth System Sciences, 25(11), 5951–5979. https://doi.org/10.5194/hess-25-5951-2021

Haider, S., Rashid, M., Tariq, M. A. U. R., & Nadeem, A. (2024a). The role of artificial intelligence (AI) and Chatgpt in water resources, including its potential benefits and associated challenges. Discover Water, 4(1), 113. https://doi.org/10.1007/s43832-024-00173-y

Haider, S., Rashid, M., Tariq, M. A. U. R., & Nadeem, A. (2024b). The role of artificial intelligence (AI) and Chatgpt in water resources, including its potential benefits and associated challenges. Discover Water, 4(1), 113. https://doi.org/10.1007/s43832-024-00173-y

Hazra, T., Anjaria, K., Bajpai, A., & Kumari, A. (2024). Introduction. In T. Hazra, K. Anjaria, A. Bajpai, & A. Kumari (Eds.), Applications of Game Theory in Deep Learning (pp. 1–12). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-54653-2_1

Hossen, M. A., Connor, J., & Ahammed, F. (2023). How to Resolve Transboundary River Water Sharing Disputes. Water, 15(14), Article 14. https://doi.org/10.3390/w15142630

Hu, S., Gao, J., & Zhong, D. (2023). Multi-agent reinforcement learning framework for real-time scheduling of pump and valve in water distribution networks. Water Supply, 23(7), 2833– 2846. https://doi.org/10.2166/ws.2023.163

Hung, F., Son, K., & Yang, Y.-C. (2021). Exploring human impacts on water scarcity uncertainty in a non-stationary environment: A Colorado River Basin case study. 2021, H41H-04. AGU Fall Meeting Abstracts.

IIJD. (n.d.). Africa's Waning Waters: Dispute over the Nile – IIJD. Retrieved March 14, 2025, from https://www.iijd.org/2021/05/31/africas-waning-waters-dispute-over-the-nile/

Jafroudi, M. (2020). A legal obligation to adapt transboundary water agreements to climate change? Water Policy, 22(5), 717–732. https://doi.org/10.2166/wp.2020.212

Janjua, S., An-Vo, D.-A., Reardon-Smith, K., & Mushtaq, S. (2025). A Three-stage Cooperative Game Model for Water Resource Allocation Under Scarcity Using Bankruptcy Rules, Nash Bargaining Solution and TOPSIS. Water Resources Management. https://doi.org/10.1007/s11269-025-04123-8

JDG Frame, Clifford-Holmes, J., & Winter, K. (2018). The Applicaiton of Stragetic Adaptive Management in the Water Services Sector.

Kairon, P., Thapliyal, K., Srikanth, R., & Pathak, A. (2020). Noisy three-player dilemma game: Robustness of the quantum advantage. Quantum Information Processing, 19(9), 327. https://doi.org/10.1007/s11128-020-02830-2

Kandeel, A. (2020, July 10). Nile Basin's GERD dispute creates risks for Egypt, Sudan, and beyond. Atlantic Council. https://www.atlanticcouncil.org/blogs/menasource/nile-basinsgerd-dispute-creates-risks-for-egypt-sudan-and-beyond/

Kansara, P., & Lakshmi, V. (2022). Water Levels in the Major Reservoirs of the Nile River Basin — A Comparison of SENTINEL with Satellite Altimetry Data. Remote Sensing, 14(18), 4667. https://doi.org/10.3390/rs14184667

Kontak, J. (2024). Measuring the Performance of Multi-Objective Reinforcement Learning algorithms — Nile River Case Study. https://repository.tudelft.nl/record/uuid:fde67c6dc57b-407d-8fe9-7ee093fb614a

Leibo, J. Z., Zambaldi, V., Lanctot, M., Marecki, J., & Graepel, T. (2017). Multi-agent Reinforcement Learning in Sequential Social Dilemmas (No. arXiv:1702.03037). arXiv. https://doi.org/10.48550/arXiv.1702.03037

Li, C., Huang, W., & Ma, Z. (2014). An Automated Negotiation Protocol Based On Game Theory. Computational Water, Energy, and Environmental Engineering. https://www.scirp.org/journal/paperinformation?paperid=48051

Loodin, N., Eckstein, G., Singh, V. P., & Sanchez, R. (2023). The Role of Data Sharing in Transboundary Waterways: The Case of the Helmand River Basin. In K. Szálkai & M.

Szalai (Eds.), Theorizing Transboundary Waters in International Relations (pp. 165–194). Springer International Publishing. https://doi.org/10.1007/978-3-031-43376-4_10

Madani, K. (2010). Game theory and water resources. Journal of Hydrology, 381(3), 225–238. https://doi.org/10.1016/j.jhydrol.2009.11.045

Mahajan, A. (2025). Comparative Analysis of Multi-Agent Reinforcement Learning Policies for Crop Planning Decision Support. https://arxiv.org/html/2412.02057

Maru, M. T. (2020). The Nile Rivalry and Its Peace and Security Implications: What Can the African Union Do? Institute for Peace and Security Studies: Addis Ababa University, 1(1).

Marwala, T. (2023a). Artificial Intelligence, Game Theory and Mechanism Design in Politics. Springer Nature. https://doi.org/10.1007/978-981-99-5103-1

Marwala, T. (2023b). Introduction to Artificial Intelligence, Game Theory, and Mechanism Design in Politics. In T. Marwala (Ed.), Artificial Intelligence, Game Theory and Mechanism Design in Politics (pp. 1–10). Springer Nature. https://doi.org/10.1007/978-981-99-5103-1_1

Mason, K., Mannion, P., & Duggan, J. (2016). Applying Multi-Agent Reinforcement Learning to Watershed Management. ResearchGate. https://www.researchgate.net/publication/299416955_Applying_MultiAgent_Reinforcement_Learning_to_Watershed_Management

McCreary, T., & Lamb, V. (2014). A political ecology of sovereignty in practice and on the map: The technicalities of law, participatory mapping, and environmental governance. Leiden Journal of International Law, 27(3). https://www.scopus.com/record/display.uri?eid=2-s2.0-84904967248&origin=inward&txGid=87fad5c0000cb1fa1ef20e0eb047cab3

McKinney, D. C. (2007). Cooperative Game Theory for Transboundary River Basins: The Syr Darya Basin. https://doi.org/10.1061/40927(243)221

Mehrparvar, M., Ahmadi, A., & Safavi, H. R. (2020). Resolving water allocation conflicts using WEAP simulation model and non-cooperative game theory. SIMULATION, 96(1), 17–30. https://doi.org/10.1177/0037549719844827

Melesse, M. (2024). Hydrology and Droughts in the Nile: A Review of Key Findings and Implications.https://www.mdpi.com/2073-4441/16/17/2521

Mirzaei-Nodoushan, F., Bozorg-Haddad, O., & Loáiciga, H. A. (2022). Evaluation of cooperative and non-cooperative game theoretic approaches for water allocation of transboundary rivers. Scientific Reports, 12(1), 3991. https://doi.org/10.1038/s41598-022-07971-1

Mostafaei, H., Kordnoori, S., Ostadrahimi, M., & Banihashemi, S. S. A. (2025). Applications of artificial intelligence in global diplomacy: A review of research and practical models. Sustainable Futures, 9, 100486. https://doi.org/10.1016/j.sftr.2025.100486

Nagheeby, M., & Amezaga, J. (2023). Decolonising water diplomacy and conflict transformation: From security-peace to equity-identity. Water Policy, 25(8), 835–850. https://doi.org/10.2166/wp.2023.043

Nash, J. (1951). Non-Cooperative Games. Annals of Mathematics, 54(2), 286–295. https://doi.org/10.2307/1969529

Nazari, S., Ahmadi, A., Kamrani Rad, S., & Ebrahimi, B. (2020). Application of non-cooperative dynamic game theory for groundwater conflict resolution. Journal of Environmental Management, 270, 110889. https://doi.org/10.1016/j.jenvman.2020.110889

Nigatu, B. A., & Lidetie, Y. L. (2025). Sovereignty vs-a-vis survival: A critical discourse analysis of the BBC and Al-Jazeera's reporting on the Great Ethiopian Renaissance Dam negotiations. Cogent Arts & Humanities, 12(1), 2451486. https://doi.org/10.1080/23311983.2025.2451486

Niklas, J. (2025). When Brussels watch from the sky: Negotiating sovereignty over data and forest in Europe. Environmental Science & Policy, 167, 104033.

https://doi.org/10.1016/j.envsci.2025.104033

Okolo, C. (2023). AI in the Global South: Opportunities and challenges towards more inclusive governance. Brookings. https://www.brookings.edu/articles/ai-in-the-global-southopportunities-and-challenges-towards-more-inclusive-governance/

Osikena, J., & David, T. (2010). Tackling the world water crisis: Reshaping the future of foreign policy. The Foreign Policy Centre. https://fpc.org.uk/publications/tackling-the-worldwater-crisis-reshaping-the-future-of-foreign-policy/

Otamendi, U., Maiza, M., Olaizola, I. G., Sierra, B., Flores, M., & Quartulli, M. (2024). Integrated Water Resource Management in the Segura Hydrographic Basin: An Artificial Intelligence Approach. Journal of Environmental Management, 370, 122526. https://doi.org/10.1016/j.jenvman.2024.122526

OWP. (2020, October 14). Nile River Conflicts. The Organization for World Peace. https://theowp.org/crisis_index/nile-river-conflicts/

Pasupuleti, M. (2024). Strategic AI: Game Theory and Reinforcement Learning for Global Diplomacy. https://www.researchgate.net/publication/385739370_Strategic_AI_Game_Theory_and_Reinforcement_Learning_for_Global_Diplomacy

Peleg, B., & Sudhölter, P. (Eds.). (2007). Introduction. In Introduction to the Theory of Cooperative Games (pp. 1–6). Springer. https://doi.org/10.1007/978-3-540-72945-7_1

RA. (2023, September 28). The Nile's influence on Sudanese culture and society. Radar Africa. https://radarafrica.com/africa/the-niles-influence-o n-sudanese-culture-and-society/

Rieu-Clarke, A., Moynihan, R., & Magsig, B.-O. (2012). UN Watercourses Convention: User's guide. IHP-HELP Centre for Water Law, Policy and Science (under the auspices of UNESCO).

Ross, D. (2024). Game Theory. In E. N. Zalta & U. Nodelman (Eds.), The Stanford Encyclopedia of Philosophy (Winter 2024). Metaphysics Research Lab, Stanford University. https://plato.stanford.edu/archives/win2024/entries/game-theory/

Sarkar, S., & Jha, P. K. (2025). Application of AI/ML in Water Resource Management to Resolve Transboundary Water Conflict. In A. Nanda, P. K. Gupta, V. Gupta, P. K. Jha, & S. K. Dubey (Eds.), Navigating the Nexus: Hydrology, Agriculture, Pollution and Climate Change, Volume 1 (pp. 431–455). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-76532-2_18

Schelble, B., Flathmann, C., Canonico, L.-B., & Mcneese, N. (2021). Understanding Human-AI Cooperation Through Game-Theory and Reinforcement Learning Models. Hawaii International Conference on System Sciences. https://doi.org/10.24251/HICSS.2021.041

Schrijver, N. J. (2021). State sovereignty in the planetary management of natural resources. Environmental Policy and Law, 51(1–2), 13–20. Scopus. https://doi.org/10.3233/EPL219002

Seyidova, I., & Gojayev, S. (2023a). Applications of Artificial Intelligence in Game Theory (SSRN Scholarly Paper No. 4664772). Social Science Research Network. https://doi.org/10.2139/ssrn.4664772

Seyidova, I., & Gojayev, S. (2023b). Applications of Artificial Intelligence in Game

Theory (SSRN Scholarly Paper No. 4664772). Social Science Research Network. https://doi.org/10.2139/ssrn.4664772

Srivastava, S., & Bullock, J. (2024). AI, Global Governance, and Digital Sovereignty (SSRN Scholarly Paper No. 4996387). Social Science Research Network. https://doi.org/10.2139/ssrn.4996387

Strasser, R. (2024, February 12). How Much Water Does Egypt Get From The Nile River — Aboutriver.com. https://www.aboutriver.com/how-much-water-does-egypt-get-from-thenile-river/ Swain, A. (2011). Challenges for water sharing in the Nile basin: Changing geo-politics and changing climate. Hydrological Sciences Journal, 56(4), 687–702. https://doi.org/10.1080/02626667.2011.577037

Tarhule, A., & Hasan, E. (2020, November 4). Satellite data provides fresh insights into the amount of water in the Nile basin. The Conversation. http://theconversation.com/satellitedata-provides-fresh-insights-into-the-amount-of-water-in-the-nile-basin-148545

Tayia, A. (2019). Transboundary Water Conflict Resolution Mechanisms: Substitutes or Complements. Water, 11(7), Article 7. https://doi.org/10.3390/w11071337

The Impact of Technology on Conflict Resolution. (2023, May 3). Join The Collective. https://www.jointhecollective.com/article/the-integration-of-technology-in-conflictresolution--past-isolation-vs--future-connectivity/

The Sovereignty Paradox. (2013). Hertie School. https://www.hertieschool.org/en/news/allcontent/detail/content/the-sovereignty-paradox-states-fight-todaysproblems-using-yesterdays-concepts

UN Water. (n.d.). Transboundary Waters: Sharing Benefits, Sharing Responsibilities — UN-Water [2008]. Retrieved February 27, 2025, from https://www.unwater.org/publications/transboundary-waters-sharing-benefits-sharing responsibilities

Vassar Labs. (2024). (4) AI-Powered Water Management — Narmada River Basin Case Study — LinkedIn. https://www.linkedin.com/pulse/ai-powered-water-management-narmada-riverbasin-case-study-anize/

Whittington, D. (2024). The long shadow of the 1959 Nile Waters Agreement. Water Policy, 26(9), 859–874. https://doi.org/10.2166/wp.2024.035

Woods, D. (2024). The Sponge Cake Dilemma over the Nile: Achieving Fairness in Resource Allocation with Cake Cutting Algorithms (No. arXiv:2310.11472). arXiv. https://doi.org/10.48550/arXiv.2310.11472

World Bank. (2019). The Nile story: 15 years of Nile cooperation - making an impact [Text/HTML]. World Bank. https://documents.worldbank.org/en/publication/documentsreports/documentdetail/en/541381468185966451

World Economic Forum. (2024). Navigating the AI Frontier: A Primer on the Evolution and Impact of AI Agents. World Economic Forum. https://reports.weforum.org/docs/WEF_Navigating_the_AI_Frontier_2024.pdf

Wu, X., He, W., & Yuan, L. (2022). Frontiers — Two-stage water resources allocation negotiation model for transboundary rivers under scarcity. https://www.frontiersin.org/journals/environmentalscience/articles/10.3389/fenvs.2022.900854/full

Wu, X., & Whittington, D. (2006). Incentive compatibility and conflict resolution in international river basins: A case study of the Nile Basin. Water Resources Research, 42(2). https://doi.org/10.1029/2005WR004238

Yao, J.-D., Hao, W.-B., Meng, Z.-G., Xie, B., Chen, J.-H., & Wei, J.-Q. (2025). Adaptive multiagent reinforcement learning for dynamic pricing and distributed energy management in virtual power plant networks. Journal of Electronic Science and Technology, 23(1), 100290. https://doi.org/10.1016/j.jnlest.2024.100290

Yibeltal, K. (2024, September 9). Gerd: Ethiopia hits out at Egypt as Nile dam row escalates. BBC News. https://www.bbc.com/news/articles/cp3dgx36gn5o

Yilma, K. (2023). In Search for a Role: The African Union and Digital Policies in Africa — Digital Society. https://link.springer.com/article/10.1007/s44206-023-00047-1

Yimer, M. (2015). THE NILE HYDRO POLITICS; A HISTORIC POWER SHIFT.

Yost, N. (n.d.). Courts or Compromise? Resolving the Nile River Dispute Under International Law – American University International Law Review. Retrieved March 27, 2025, from https://auilr.org/2025/03/24/courts-or-compromise-resolving-the-nile-river-disputeunder-international-law/

Zedan, B. (2018). NILE WATER CONFLICT MANAGEMENT.

Zeitoun, M., & Warner, J. (2006). Hydro-Hegemony- a Framework for Analysis of TransBoundary Water Conflicts. Water Policy 8 (2006) 5, 8. https://doi.org/10.2166/wp.2006.054

Zeng, Y., Li, J., Cai, Y., Tan, Q., & Dai, C. (2019). A hybrid game theory and mathematical programming model for solving trans-boundary water conflicts. Journal of Hydrology, 570, 666–681. https://doi.org/10.1016/j.jhydrol.2018.12.053

Zhao, T., Song, C., Yu, J., Xing, L., Xu, F., Li, W., & Wang, Z. (2025). Leveraging Immersive Digital Twins and AI-Driven Decision Support Systems for Sustainable Water Reserves Management: A Conceptual Framework. Sustainability, 17(8), Article 8. https://doi.org/10.3390/su17083754

Zomorodian, M., Lai, S. H., Homayounfar, M., Ibrahim, S., & Pender, G. (2017). Development and application of coupled system dynamics and game theory: A dynamic water conflict resolution method. PLOS ONE, 12(12), e0188489. https://doi.org/10.1371/journal.pone.01884